Assignment 5

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1.

$$P(J|I) = \frac{P(I,J)}{P(I)} = \frac{1}{Z'} e^{\frac{-E(I,J)}{T}}$$

$$E(I,J) = c \sum_{\alpha} (I_\alpha - J_\alpha)^2 + \sum_{\{\alpha\beta\}(J_\alpha - J_\beta)^2}$$

$$\begin{aligned} p &= P(J_\alpha = 1 | \alpha J) \\ &= \frac{e^{\frac{-c(I_\alpha - 1)^2 - \sum_{\beta:\alpha \ \ \beta} (1 - J_\beta)^2}{T}}}{e^{\frac{-c\sum_\alpha (I_\alpha - 1)^2 - \sum_{\beta:\alpha \ \ \beta} (1 - J_\beta)^2}{T}} + e^{\frac{-c\sum_\alpha (I_\alpha)^2 - \sum_{\beta:\alpha \ \ \beta} (J_\beta)^2}{T}}} \end{aligned}$$

2.

3.

